

Logical & Physical Data Independence

Q. Define Data Independence. Explain the concepts of Logical and Physical Data Independence and highlight the key differences between them.

> **Definition to Remember**

1. Types of Data Independence

Data independence is linked to the three-level abstraction architecture and is classified into two types:

1. Physical Data Independence

2. Logical Data Independence

2. Difference: Physical vs Logical Data Independence

Feature	Physical Data Independence	Logical Data Independence
Definition	Physical data independence is the ability to change the physical storage of data without affecting the logical structure of the data.	Logical data independence is the ability to change the logical structure of the data without affecting the physical storage of the data.
Types	Physical data independence is further divided into two types: 1. Storage independence: The ability to change the physical storage of data without affecting the logical structure of the data. 2. Access independence: The ability to change the access method of data without affecting the logical structure of the data.	Logical data independence is further divided into two types: 1. Query independence: The ability to change the query language without affecting the physical storage of the data. 2. Schema independence: The ability to change the schema of the data without affecting the physical storage of the data.
Example	Example: Changing the physical storage of data from one storage device to another without affecting the logical structure of the data.	Example: Changing the query language from SQL to another query language without affecting the physical storage of the data.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**

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